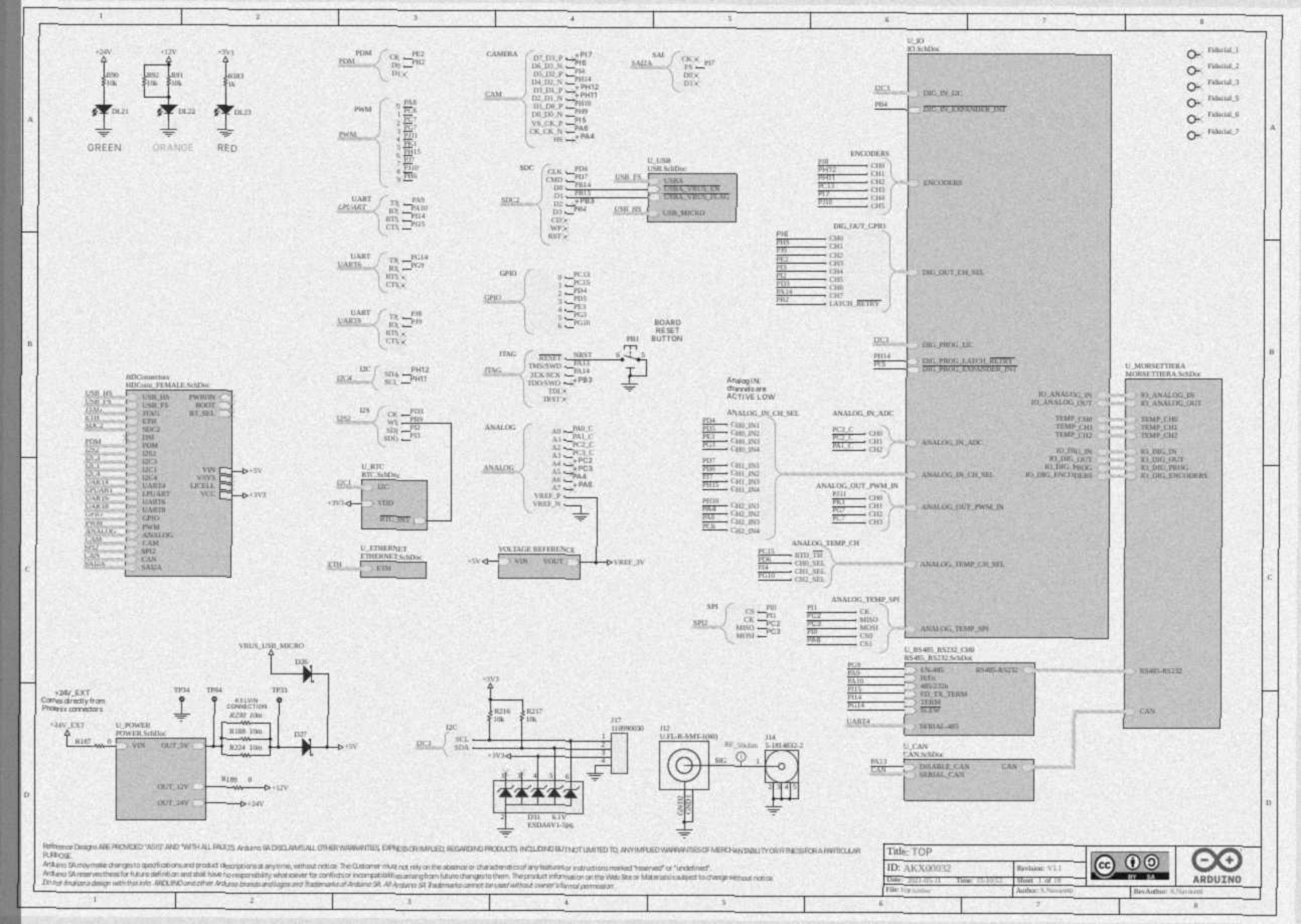
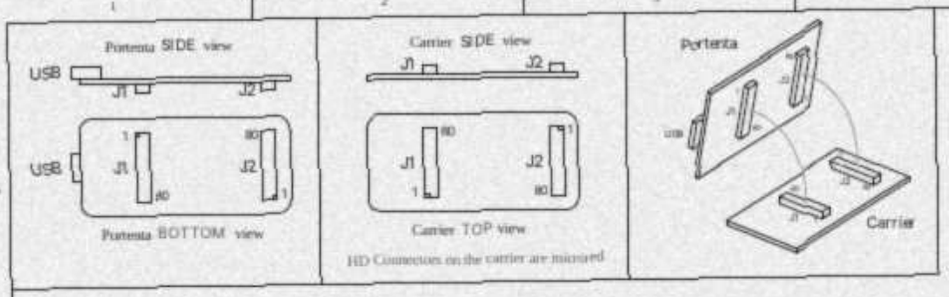

+24V
+12V
+3V3
R190 10k
R192 10k
R191 10k
R193 1k
DL21
DL22
DL23
GREEN
ORANGE
RED
PDM
PWM
UART
CAMERA
CAM
SAI
SDC
GPIO
BOARD RESET BUTTON
JTAG
I2C
I2S
ANALOG
VOLTAGE REFERENCE
U_RTC
RTC_SxDoc
U_ETHERNET
ETHERNET.SchDoc
U_USB
USB.SchDoc
U_IO
IO.SchDoc
ENCODERS
DIG_OUT_GPIO
ANALOG_IN_CH_SEL
ANALOG_IN_ADC
ANALOG_OUT_PWM_IN
ANALOG_TEMP_CH
ANALOG_TEMP_SPI
U_MORSETTIERA
MORSETTIERA.SchDoc
U_RS485_RS232_CH0
RS485_RS232.SchDoc
U_CAN
CAN.SchDoc
Fiducial_1
Fiducial_2
Fiducial_3
Fiducial_5
Fiducial_6
Fiducial_7
BDConnector
MDCom_FEMALE.SchDoc
VBUS_USB_MICRO
D36
+24V_EXT
Comes directly from Phoenix connectors
U_POWER
POWER.SchDoc
KELVIN CONNECTION
R187 0
R188 10m
R224 10m
D37
R189 0
+5V
+12V
+24V
R216 10k
R217 10k
I2C
J17
I2C
SCL
SDA
+3V3
J12
U.FL-R-SMT-1(60)
RF_50ohm
J14
5-181 4632-2
GND
GND
D31 K1V ESDA6V1-5p6
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Title: TOP
ID: AKX00032
Revision: V1.1
Date: 2021-05-11
Time: 15:10:51
Sheet 1 of 10
File: Top.SchDoc
Author: S.Navarro
Res Author: S.Navarro
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IF NECESSARY ROUTE CAMERA WITH DIFFERENTIAL PAIR RULES CHANGING THE NET NAMES. REFER TO SHARED PINS TABLE

SHARED PINS TABLE

NET	BUS 1	BUS 2
PB3	JTAG	SDC2
PH11	CAM	I2C4
PH12	CAM	I2C4
PI5	CAM	SAI2A
PI6	CAM	SAI2A
PI7	CAM	SAI2A
PA4	CAM	ANALOG
PA6	CAM	ANALOG
PC2	ANALOG	SPI2
PC3	ANALOG	SPI2

POWER NETS TABLE

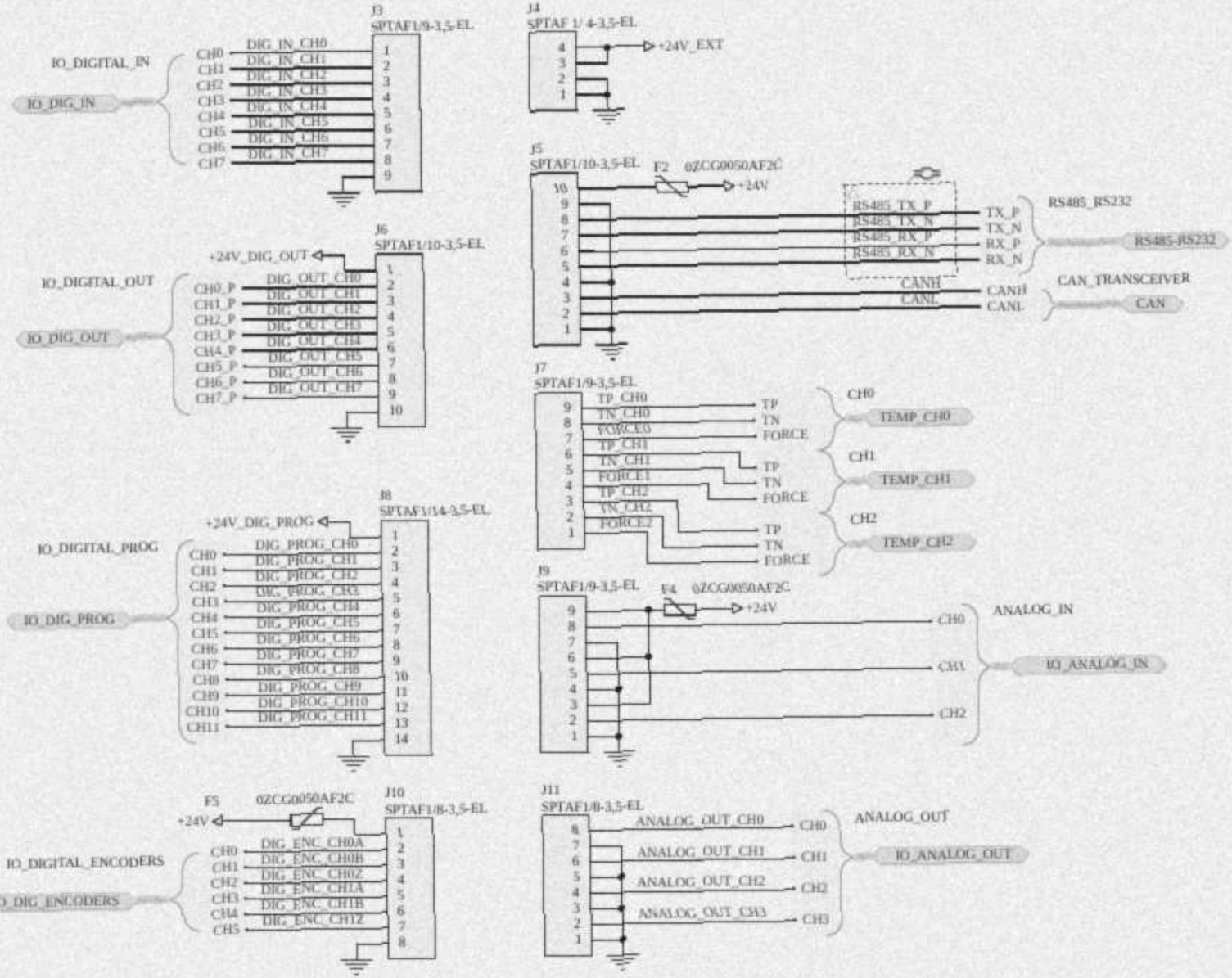
NET	TYPE	RANGES	DESCRIPTION
VIN	PORTENTA INPUT	4.1V to 6V.	Default 3.3V, PMIC (U10) programmable output.
VCC	PORTENTA OUTPUT	1.1V to 3.3V in steps, max 1A.	Default 3.3V, PMIC (U10) programmable output.
VSYS	PORTENTA BIDIRECTIONAL OUTPUT	RESERVED, DO NOT USE 3.5V to 4.2V, max 900mA.	Default 4.2V, PMIC (U10) programmable output which is also the input voltage of the bucks inside the PMIC itself.
LICELL	PORTENTA INPUT	Coin cell max 3.6V, max 46uA.	Max 4uA with PMIC (U10) in coin cell mode, max 46uA with PMIC in standby/suspend mode.

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Title:	HD Connectors female
ID:	AKX00032
Revision:	V1.1
Date:	2021-05-11
Time:	10:00:53
Sheet:	2 of 19
File:	ditron_HMBLE.SchDoc
Author:	S.Navarro
RevAuthor:	*

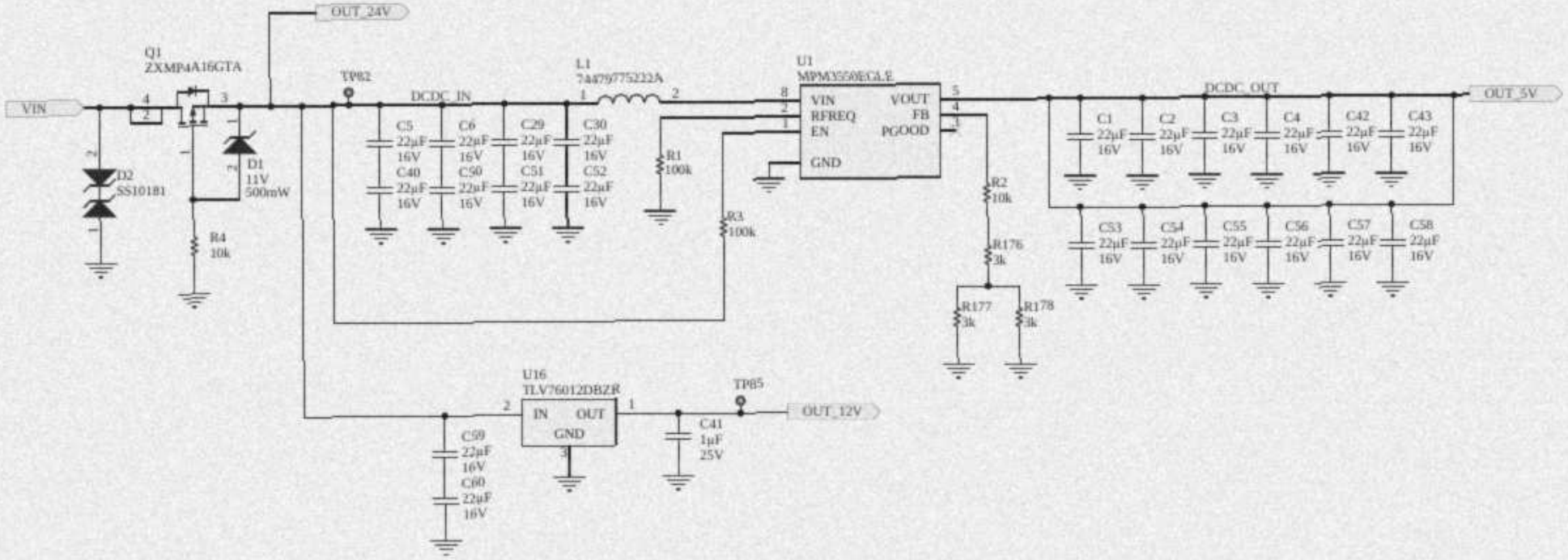
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Title: MORSETTIERA			
ID: AKX00032	Revision: V1.1		
Date: 2021-05-11 Time: 15:10:54	Sheet 3 of 19		
File: MORSETTIERA.SchDoc	Author: S.Navaretti	RevAuthor: S.Navaretti	

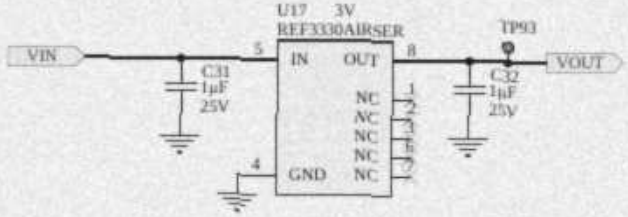


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Title: POWER			
ID: AKX00032		Revision: V1.1	
Date: 2021-05-11	Time: 15:10:50	Sheet 4 of 19	
File: POWER.SchDoc		Author: S.Navaretti	RevAuthor: S.Navaretti

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Title: VOLTAGE REFERENCE		
ID: AKX00032	Revision: V1.1	
Date: 2021-05-11 Time: 15:10:56	Sheet 5 of 19	
File: ADC_VOLTAGE_REFERENCE.SchDoc	Author: S.Navaretti	RevAuthor: S.Navaretti

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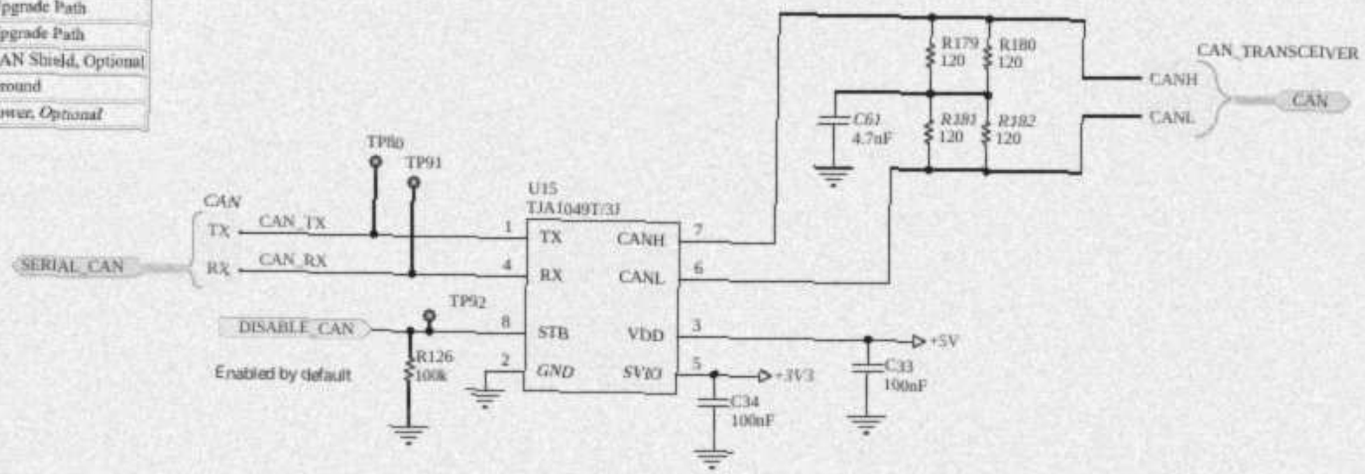
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RJ45 connector

RJ10, RJ45 CAN Bus PinOut

RJ45 Pin #	RJ10 Pin #	Signal name	Signal Description
1	2	CAN_H	Dominant High
2	3	CAN_L	Dominant Low
3	4	CAN_GND	Ground
4	-	Reserved	Upgrade Path
5	-	Reserved	Upgrade Path
6	-	CAN_SHLD	CAN Shield, Optional
7	-	CAN_GND	Ground
8	2	*CAN_V+*	*Power, Optional*





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Title: CAN			
ID: AKX00032	Revision: V1.1		
Date: 2021-05-11 Time: 15:10:57	Sheet 6 of 19		
File: CAN.SchDoc	Author: S.Navaretti	RevAuthor: S.Navaretti	

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PIN DESCRIPTIONS

Pin	Name	RS-232	RS-485 Full Duplex	RS-485 Half Duplex
1				
2	GND	Ground		
3	TXOUT, RO	Transmitter 1 Output	Y Driver Neg Output	B/Z Neg Input/Output
4	TZOUT, A/Y	Transmitter 2 Output	Y Driver Pos Output	A/Y Pos Input/Output
5				
6	R1OUT	Receiver 1 Output	X	X
7	R2OUT, RO	Receiver 2 Output	Receiver TTL Output	Receiver TTL Output
8				
9				
10	SLEW	Low slew, slow/down mode when low		
11	RS485	Data rate limited to 250Kbps when low		
12	FD_TX_TERM	X	120Ω Y-Z termination enabled when both TERM and FD_TX_TERM are high	X
13	TERM	X	120Ω A/B termination enabled when high	
14	RS-485/RS-232	H	1	1
15	HALF/FULL	X	0	1
16				
17	GND	Ground		
18	R2IN, A	Receiver 2 Input	A Pos Receiver Input	X
19	R1IN, B	Receiver 1 Input	B Neg Receiver Input	X
20	RE	X	Receiver enabled when low	
21	T2IN, DE	Transmitter 2 Input	Driver enabled when high	
22	T1IN, DI	Transmitter 1 Input	Driver TTL Input	
23				
24				
25	V-	Charge pump negative supply, 0.1µF from ground		
26	C2-	Charge pump cap 2 negative lead		
27	C2+	Charge pump cap 2 positive lead, 0.1µF		
28	V+	Charge pump positive supply, 0.1µF to ground		
29	C1+	Charge pump cap 1 positive lead, 0.1µF		
30	VL	Logic Supply for TTL Inputs and Outputs. V_L = +1.65V to +5.5V or tie to V_{CC}		
31	VCC	When Supply V_{CC} = +3.0V to +5.5V, bypass to ground with 1.0µF		
32	C1-	Charge pump cap 1 negative lead		

SHDN	INPUTS RS-485/RS-232	INPUTS RS-485/RS-232	OUTPUTS R1OUT, R2OUT/RO
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	X	X	High-Z

SHDN	INPUTS RS-485/RS-232	INPUTS HALF/FULL	OUTPUTS R1OUT, R2OUT/RO
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	X	X	High-Z

SHDN	INPUTS RS-485/RS-232	INPUTS HALF/FULL	INPUTS TE	INPUTS DE	OUTPUTS R1OUT, R2OUT/RO
0	0	0	0	0	High-Z
0	0	0	0	1	0
0	0	0	1	0	High-Z
0	0	0	1	1	1
0	0	1	0	0	High-Z
0	0	1	0	1	0
0	0	1	1	0	High-Z
0	0	1	1	1	1
0	1	0	0	0	High-Z
0	1	0	0	1	0
0	1	0	1	0	High-Z
0	1	0	1	1	1
0	1	1	0	0	High-Z
0	1	1	0	1	0
0	1	1	1	0	High-Z
0	1	1	1	1	1
1	X	X	X	X	High-Z

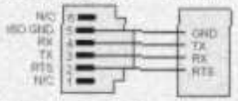
SHDN	TERM	FD_TX_TERM	RS-485/RS-232	HALF/FULL	TE	DE	R1OUT	R2OUT/RO
0	0	0	0	0	0	0	High-Z	High-Z
0	0	0	0	0	0	1	0	1
0	0	0	0	0	1	0	High-Z	High-Z
0	0	0	0	0	1	1	1	0
0	0	0	0	1	0	0	High-Z	High-Z
0	0	0	0	1	0	1	0	1
0	0	0	0	1	1	0	High-Z	High-Z
0	0	0	0	1	1	1	1	0
0	0	0	1	0	0	0	High-Z	High-Z
0	0	0	1	0	0	1	0	1
0	0	0	1	0	1	0	High-Z	High-Z
0	0	0	1	0	1	1	1	0
0	0	0	1	1	0	0	High-Z	High-Z
0	0	0	1	1	0	1	0	1
0	0	0	1	1	1	0	High-Z	High-Z
0	0	0	1	1	1	1	1	0
0	0	1	0	0	0	0	High-Z	High-Z
0	0	1	0	0	0	1	0	1
0	0	1	0	0	1	0	High-Z	High-Z
0	0	1	0	0	1	1	1	0
0	0	1	0	1	0	0	High-Z	High-Z
0	0	1	0	1	0	1	0	1
0	0	1	0	1	1	0	High-Z	High-Z
0	0	1	0	1	1	1	1	0
0	0	1	1	0	0	0	High-Z	High-Z
0	0	1	1	0	0	1	0	1
0	0	1	1	0	1	0	High-Z	High-Z
0	0	1	1	0	1	1	1	0
0	0	1	1	1	0	0	High-Z	High-Z
0	0	1	1	1	0	1	0	1
0	0	1	1	1	1	0	High-Z	High-Z
0	0	1	1	1	1	1	1	0
0	1	0	0	0	0	0	High-Z	High-Z
0	1	0	0	0	0	1	0	1
0	1	0	0	0	1	0	High-Z	High-Z
0	1	0	0	0	1	1	1	0
0	1	0	0	1	0	0	High-Z	High-Z
0	1	0	0	1	0	1	0	1
0	1	0	0	1	1	0	High-Z	High-Z
0	1	0	0	1	1	1	1	0
0	1	0	1	0	0	0	High-Z	High-Z
0	1	0	1	0	0	1	0	1
0	1	0	1	0	1	0	High-Z	High-Z
0	1	0	1	0	1	1	1	0
0	1	0	1	1	0	0	High-Z	High-Z
0	1	0	1	1	0	1	0	1
0	1	0	1	1	1	0	High-Z	High-Z
0	1	0	1	1	1	1	1	0
0	1	1	0	0	0	0	High-Z	High-Z
0	1	1	0	0	0	1	0	1
0	1	1	0	0	1	0	High-Z	High-Z
0	1	1	0	0	1	1	1	0
0	1	1	0	1	0	0	High-Z	High-Z
0	1	1	0	1	0	1	0	1
0	1	1	0	1	1	0	High-Z	High-Z
0	1	1	0	1	1	1	1	0
0	1	1	1	0	0	0	High-Z	High-Z
0	1	1	1	0	0	1	0	1
0	1	1	1	0	1	0	High-Z	High-Z
0	1	1	1	0	1	1	1	0
0	1	1	1	1	0	0	High-Z	High-Z
0	1	1	1	1	0	1	0	1
0	1	1	1	1	1	0	High-Z	High-Z
0	1	1	1	1	1	1	1	0
1	X	X	X	X	X	X	High-Z	High-Z

For DE and TE, active high enables the corresponding driver. In receiver mode, ...

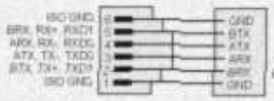
FD_TX_TERM	TERM	RS-485/RS-232	HALF/FULL	TE	TX TERM	RX TERM
Pin 12	Pin 13	Pin 14	Pin 15	Pins 8,9	Pins 6,7	Pins 10,11
X	0	1	0			
0	1	1	0			1Ω
1	1	1	0	0	1Ω	1Ω
X	0	1	1			
X	1	1	1	0	1Ω	-
X	X	0	X	X	-	-

Use DE and TE active high enables for the termination shorting in each mode.

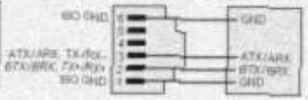
RS232 INTERFACE



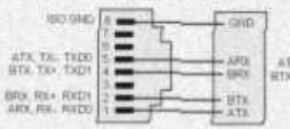
RS485, RJ11, FULL DUPLEX



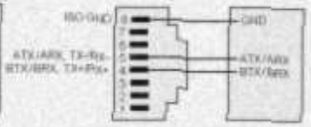
RS485, RJ11, HALF DUPLEX

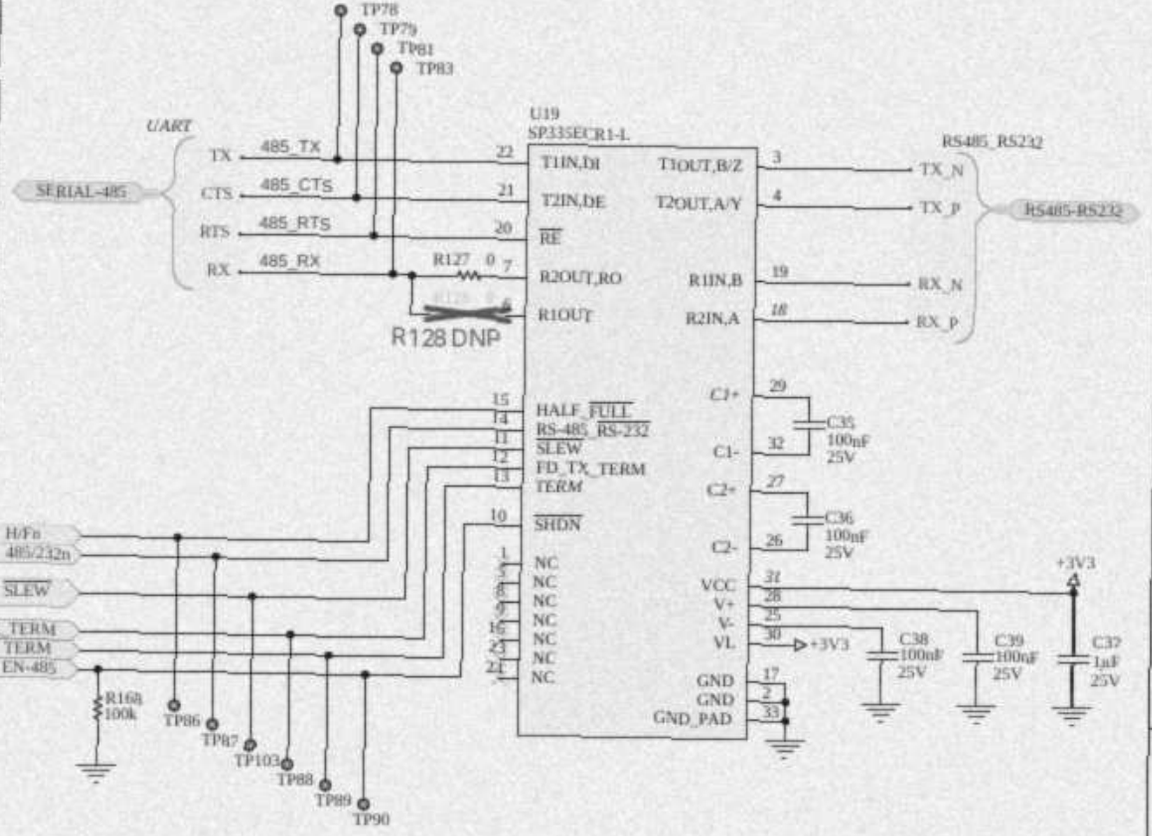


RS485, RJ45, FULL DUPLEX



RS485, RJ45, HALF DUPLEX



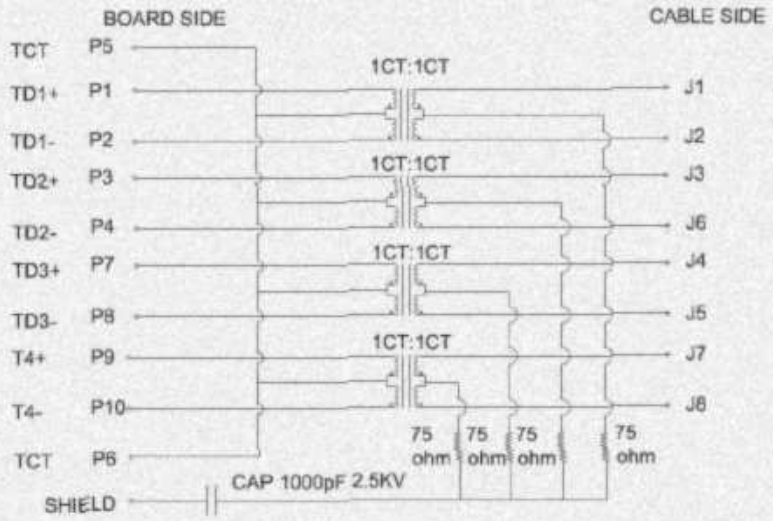


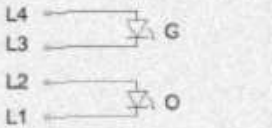
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Title: RS485 / RS422 / RS232		
ID: AKX00032	Revision: V1.1	
Date: 2021-05-11 Time: 15:10:58	Sheet 7 of 10	
File: RS485_RS232.SchDoc	Author: S.Navaretti	RevAuthor: S.Navaretti

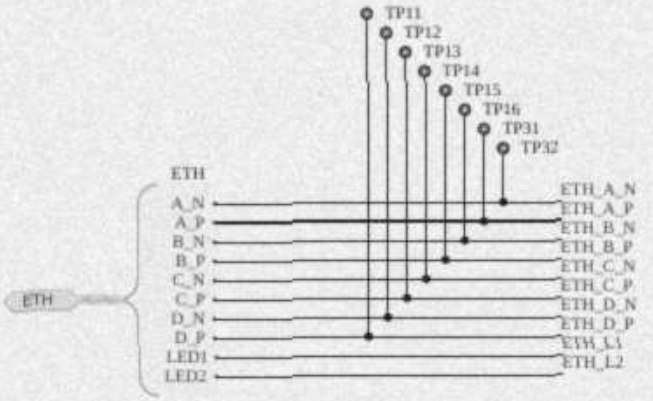
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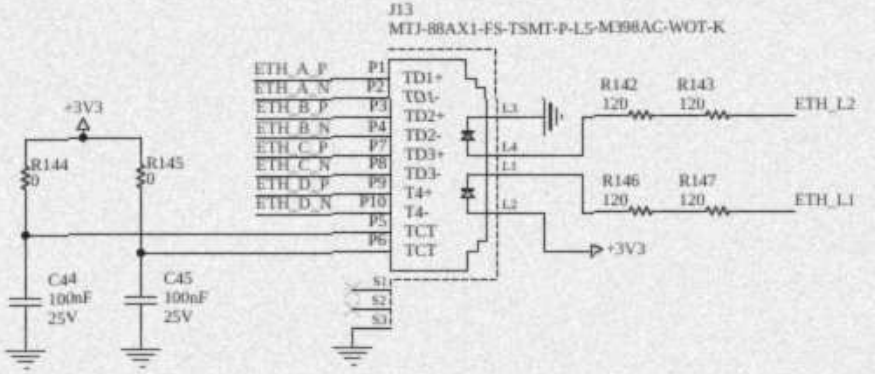
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Pin	Green	Pin	Orange
L4	+	L2	+
L3	-	L1	-



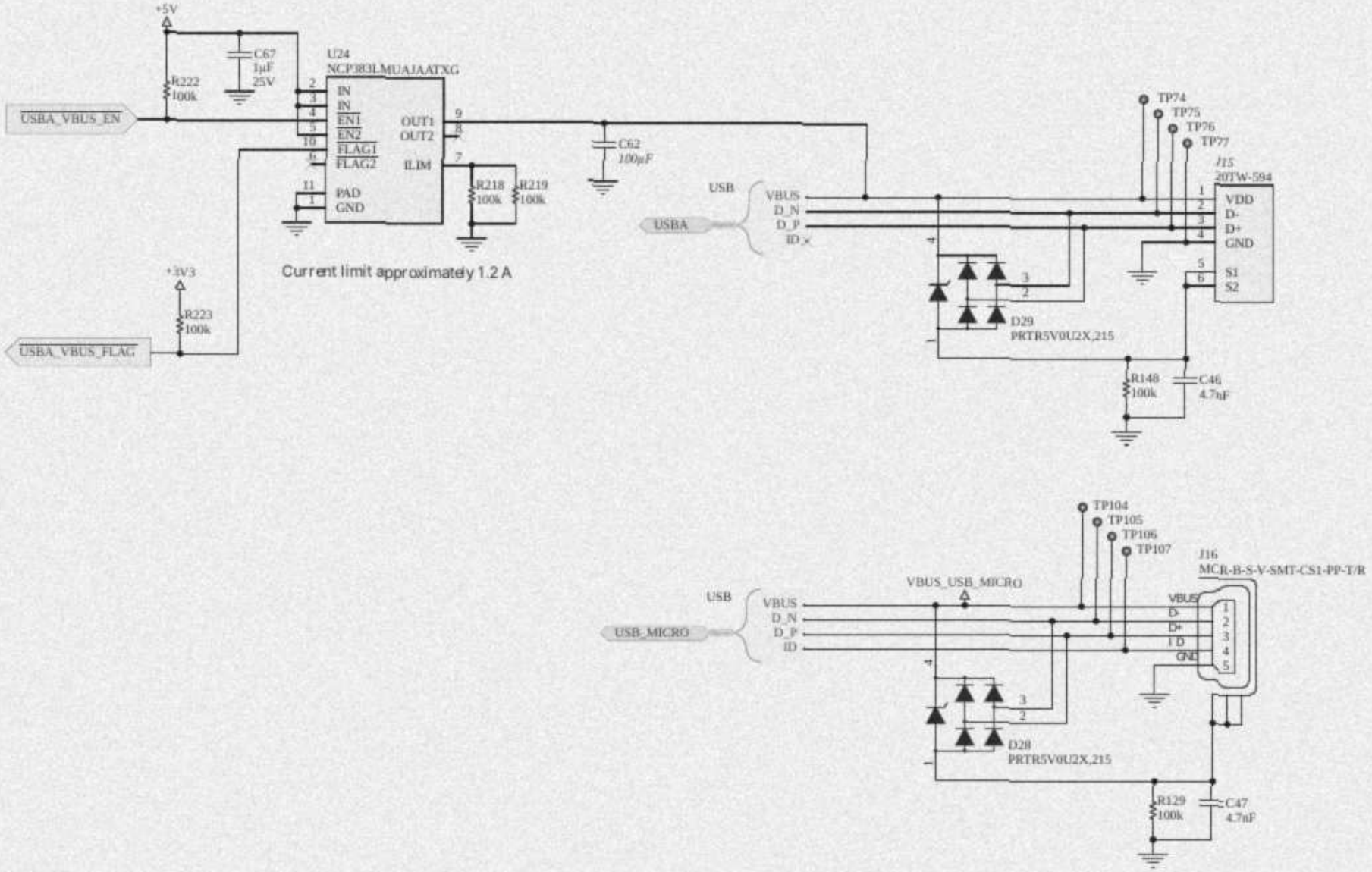


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Title: ETHERNET		
ID: AKX00032	Revision: V1.1	
Date: 2021-05-13 Time: 15:10:58	Sheet 8 of 19	
File: ETHERNET.SchDoc	Author: S.Navaretti	RevAuthor: S.Navaretti

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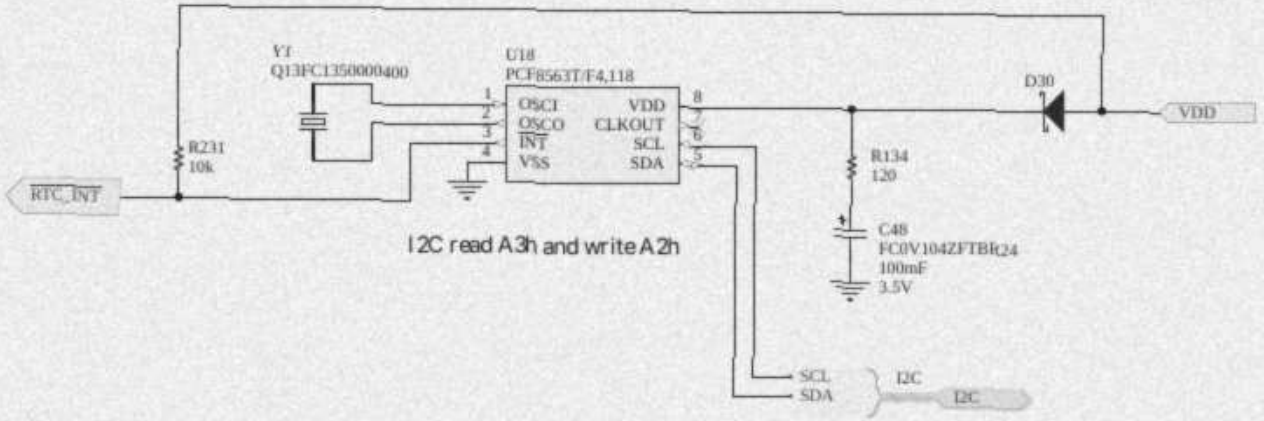


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Title: USB			
ID: AKX00032	Revision: V1.1		
Date: 2023-05-11 Time: 15:10:59	Sheet 9 of 19		
File: USB.SchDoc	Author: S.Navaretti	RevAuthor: S.Navaretti	

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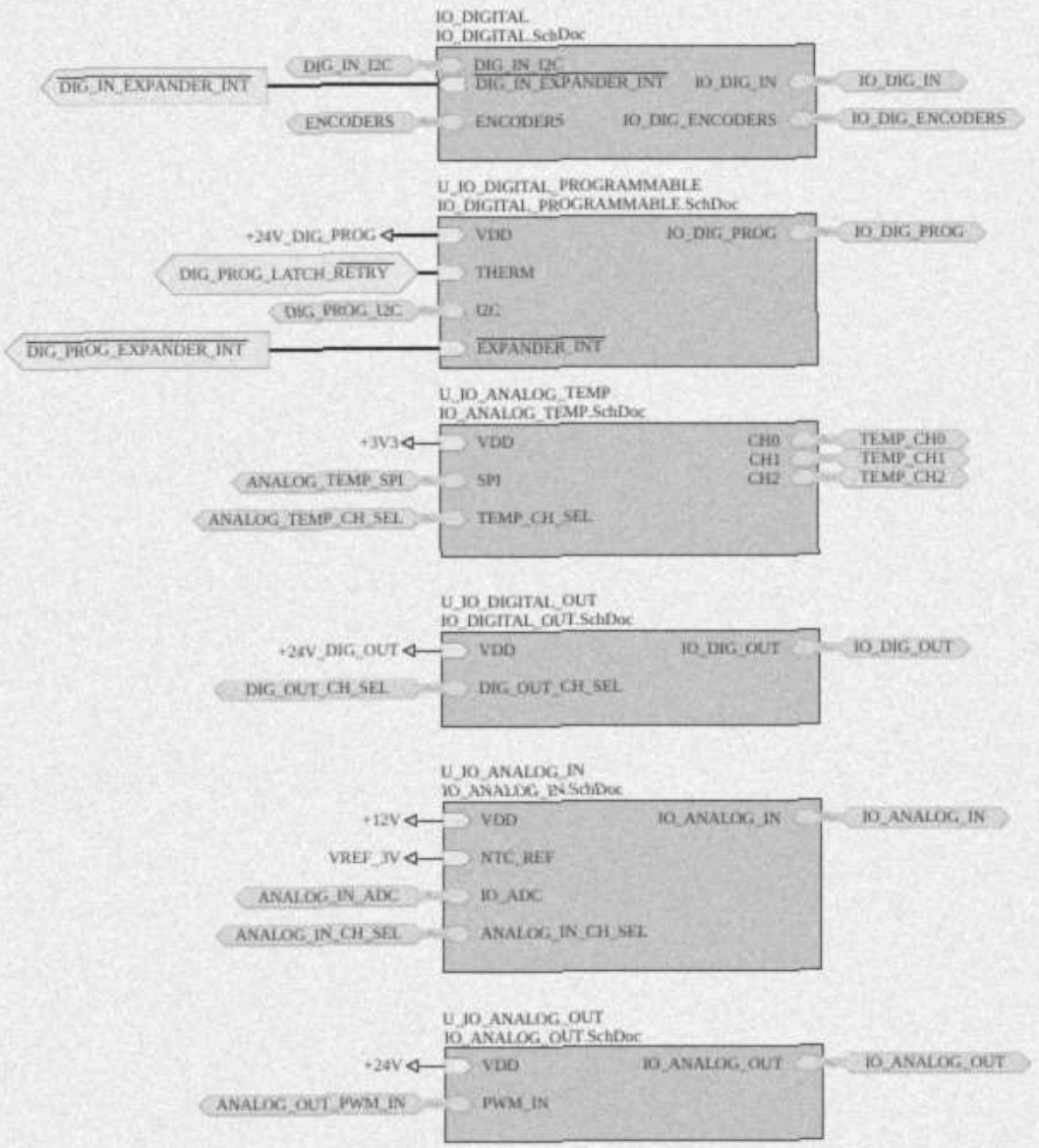


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Title: RTC			
ID: AKX00032	Revision: V1.1		
Date: 2021-05-11 Time: 15:11:00	Sheet 10 of 19		
File: RTC_SchDoc	Author: S.Navaretti	RevAuthor: S.Navaretti	

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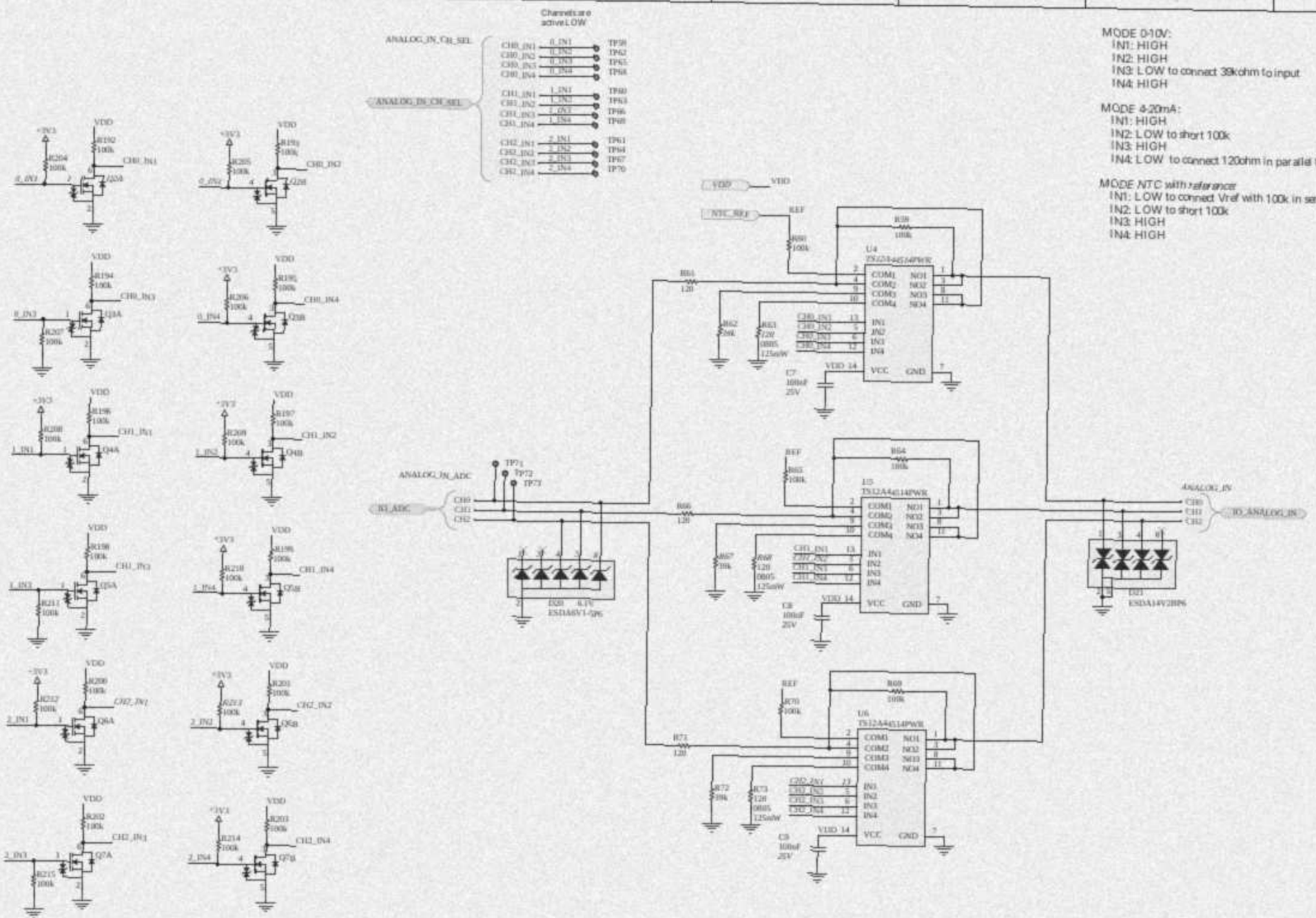


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Title: IO TOP			
ID: AKX00032		Revision: V1.1	
Date: 2021-05-11	Time: 15:11:01	Sheet 11 of 19	
File: IO.SchDoc		Author: S.Navaretti	RevAuthor: S.Navaretti

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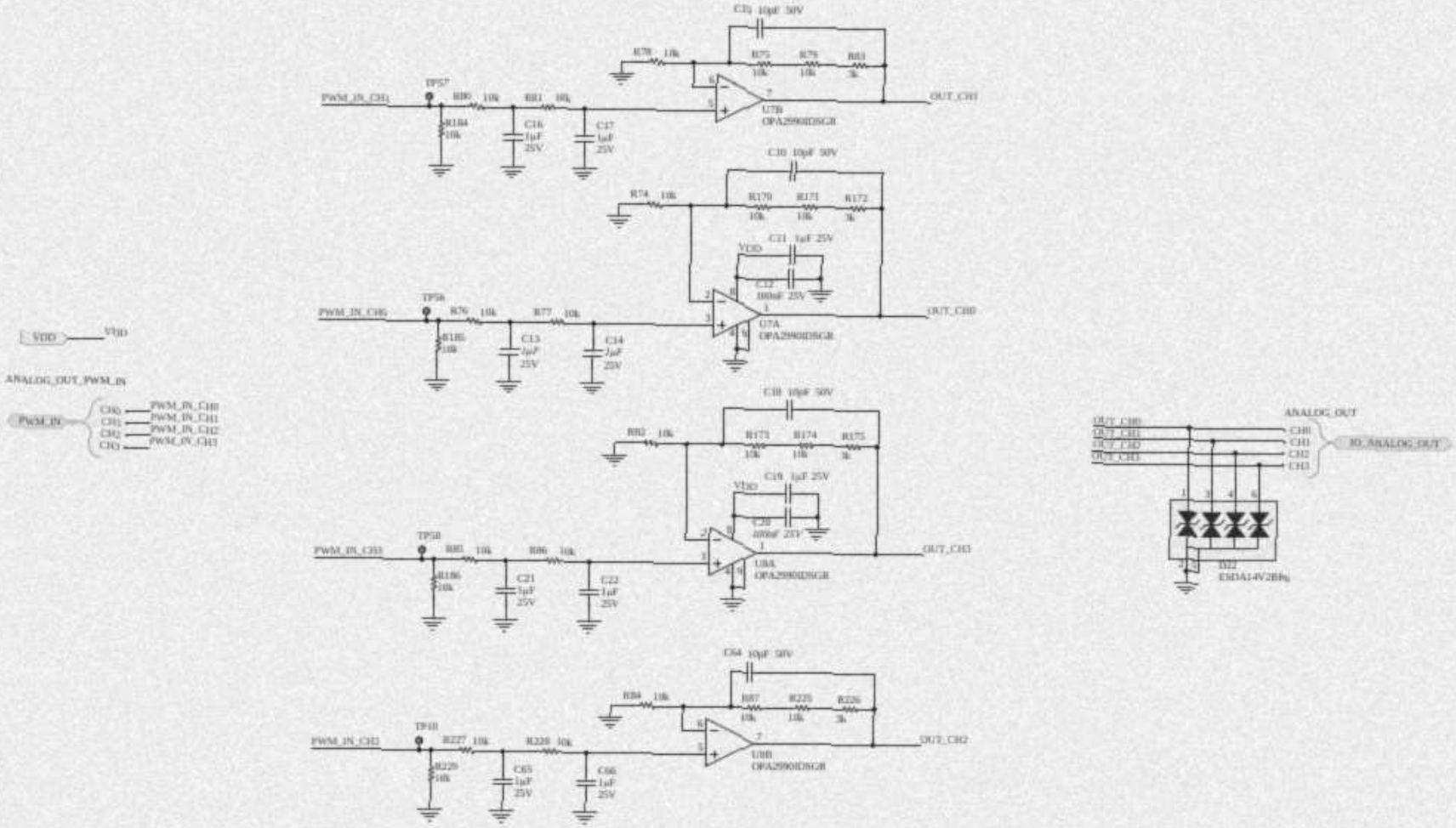


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Title:	ANALOG IN
ID:	AKX00032
Revision:	V1.1
Date:	2021-05-21 Time: 15:17:05
Sheet:	13 of 15
File:	01_ANALOG_IN.SchDoc
Author:	S.Novaretti
Rev.Author:	S.Novaretti

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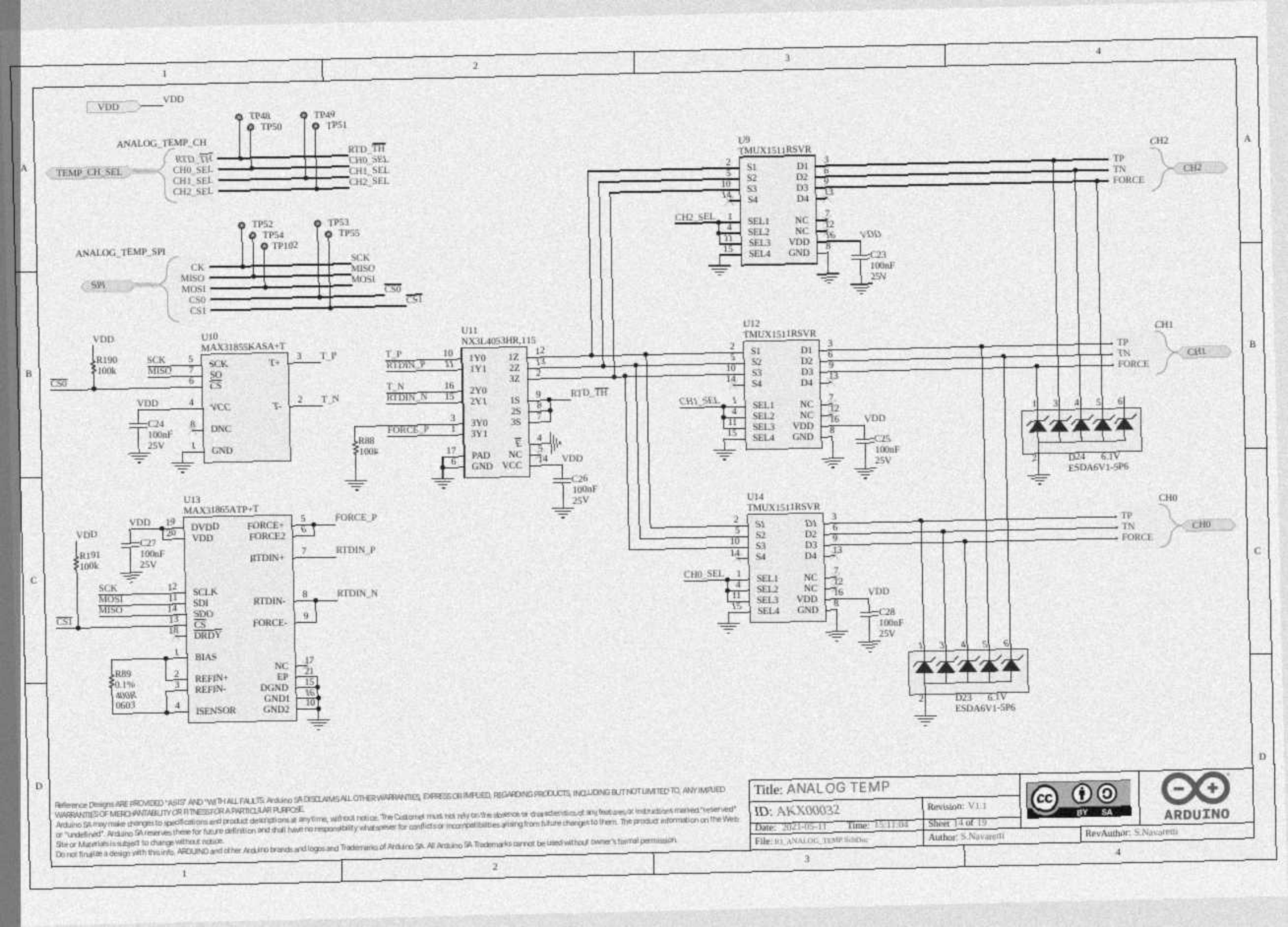


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Title:	ANALOG OUT
ID:	AKX00032
Revision:	V1.1
Date:	2021-06-11 Time: 15:31:05
Sheet:	13 of 18
File:	IO_ANALOG_OUT.SchDoc
Author:	S.Savarese
Rev.Author:	S.Savarese

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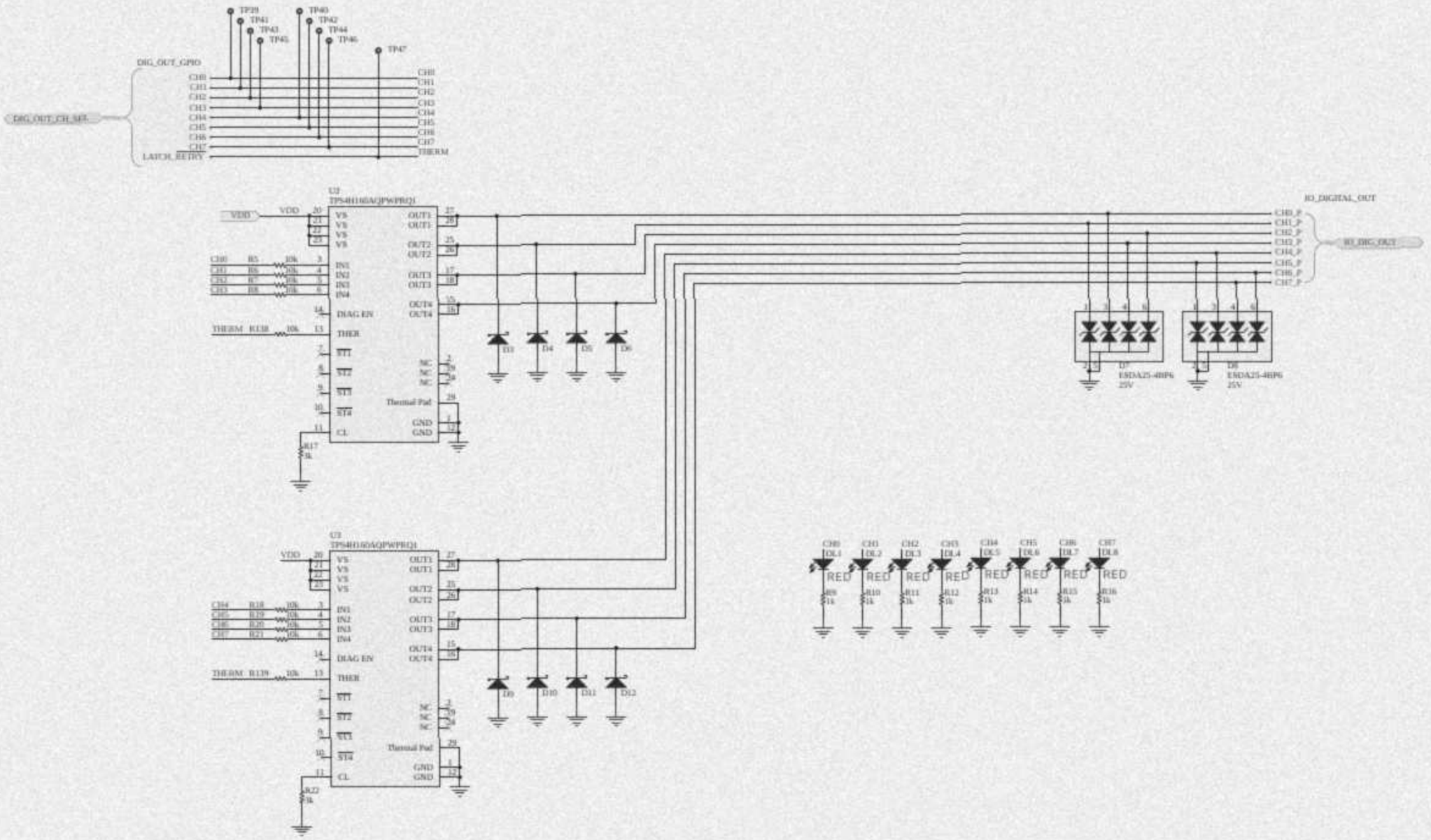


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Title: ANALOG TEMP	
ID: AKX00032	Revision: V1.1
Date: 2021-05-11 Time: 15:11:04	Sheet 14 of 19
File: 01_ANALOG_TEMP.SchDoc	Author: S.Navaretti RevAuthor: S.Navaretti

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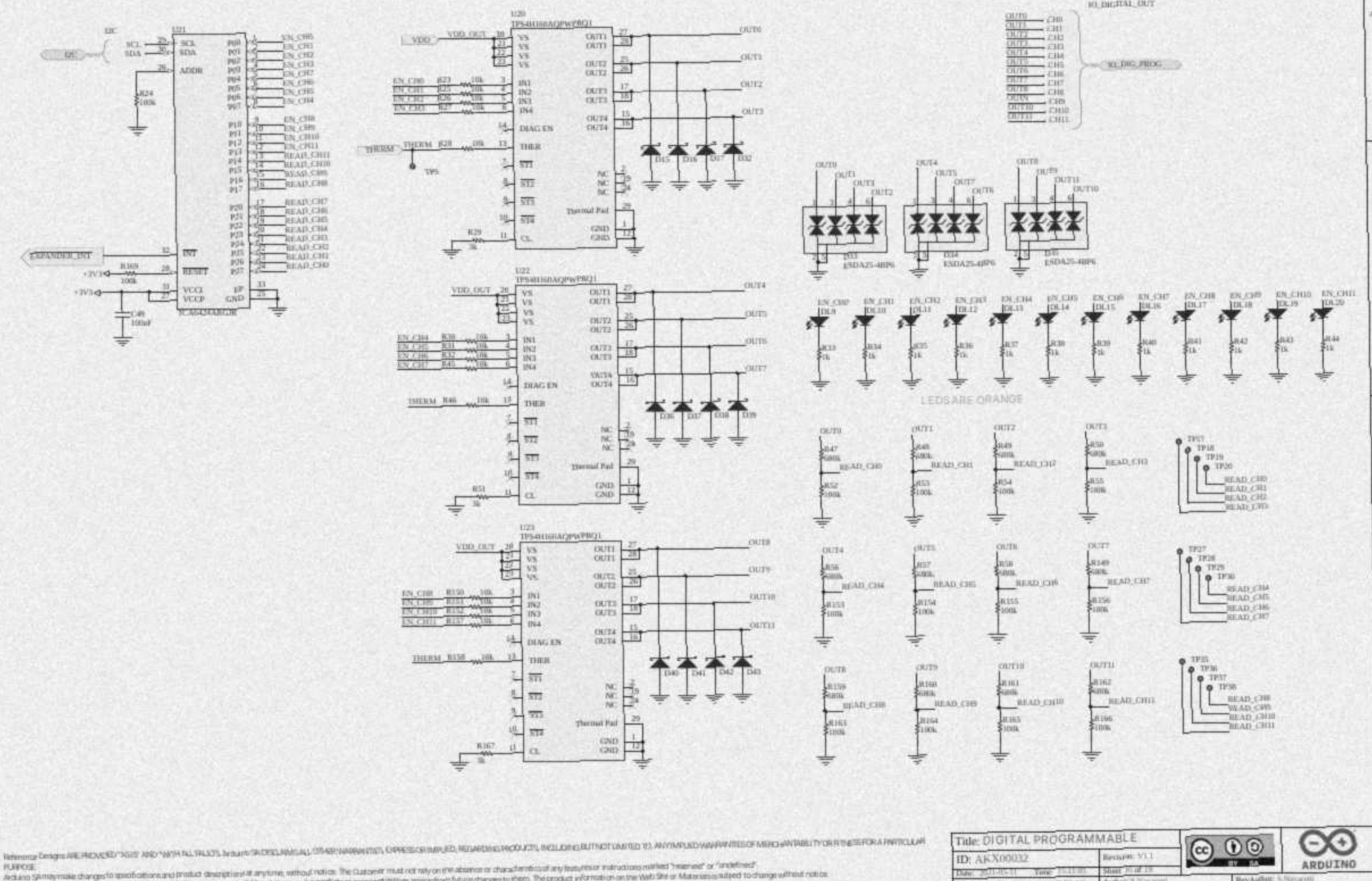


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Title: DIGITAL OUT			
ID: AKX00032		Revision: V1.1	
Date: 2023-06-13	Time: 10:13:04	Sheet: 11 of 19	
File: 03_DIGITAL_OUT.SchDoc		Author: S.Nastasi	Rev Author: S.Nastasi

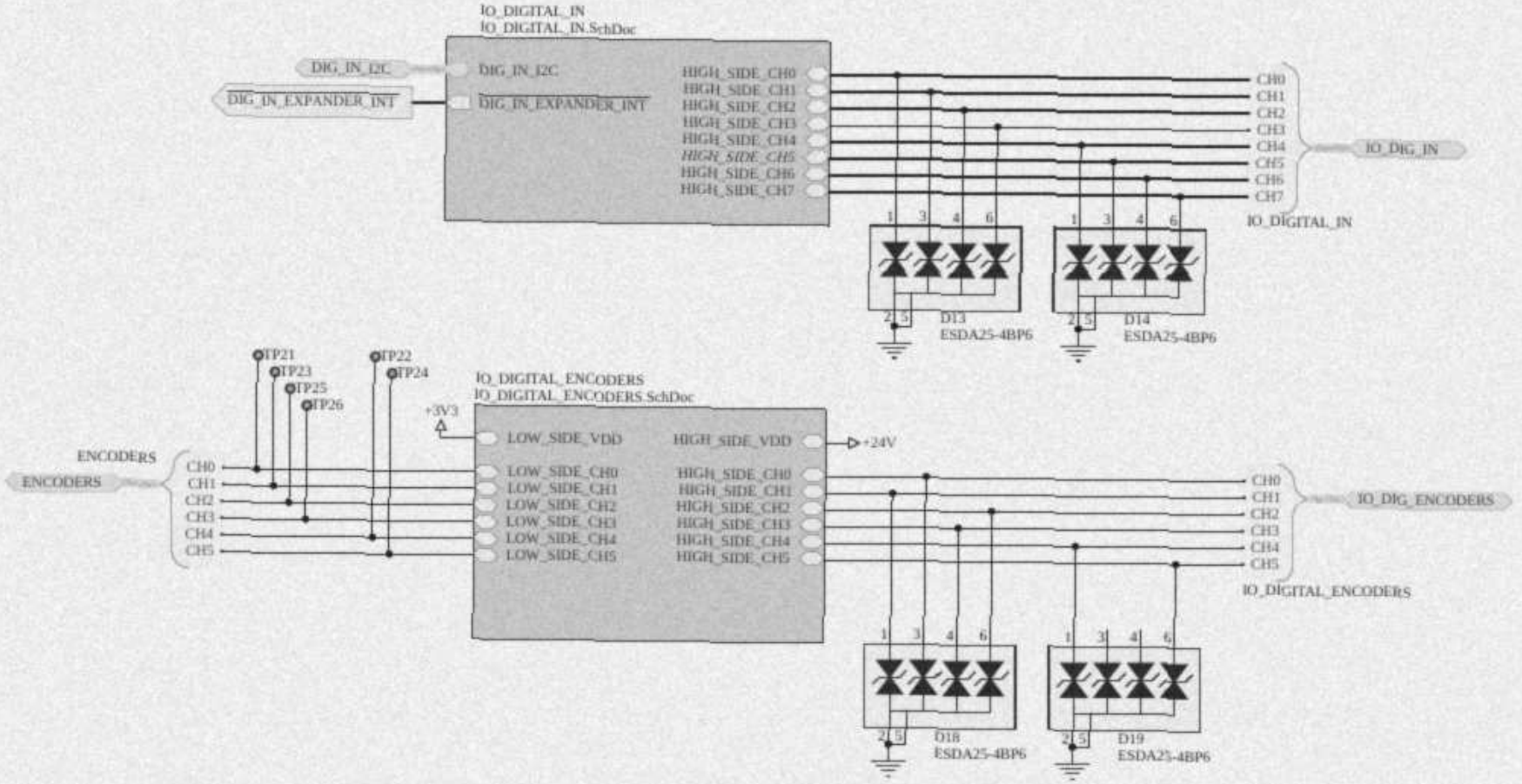
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Title: DIGITAL PROGRAMMABLE			
ID: AKX00032		Rev/ver: V1.1	
Date: 2021-05-11	Time: 11:11:00	Sheet: 6 of 12	
File: 11_DIGITAL_PROGRAMMABLE.SchDoc		Author: S.Navarotti	Rev Author: S.Navarotti



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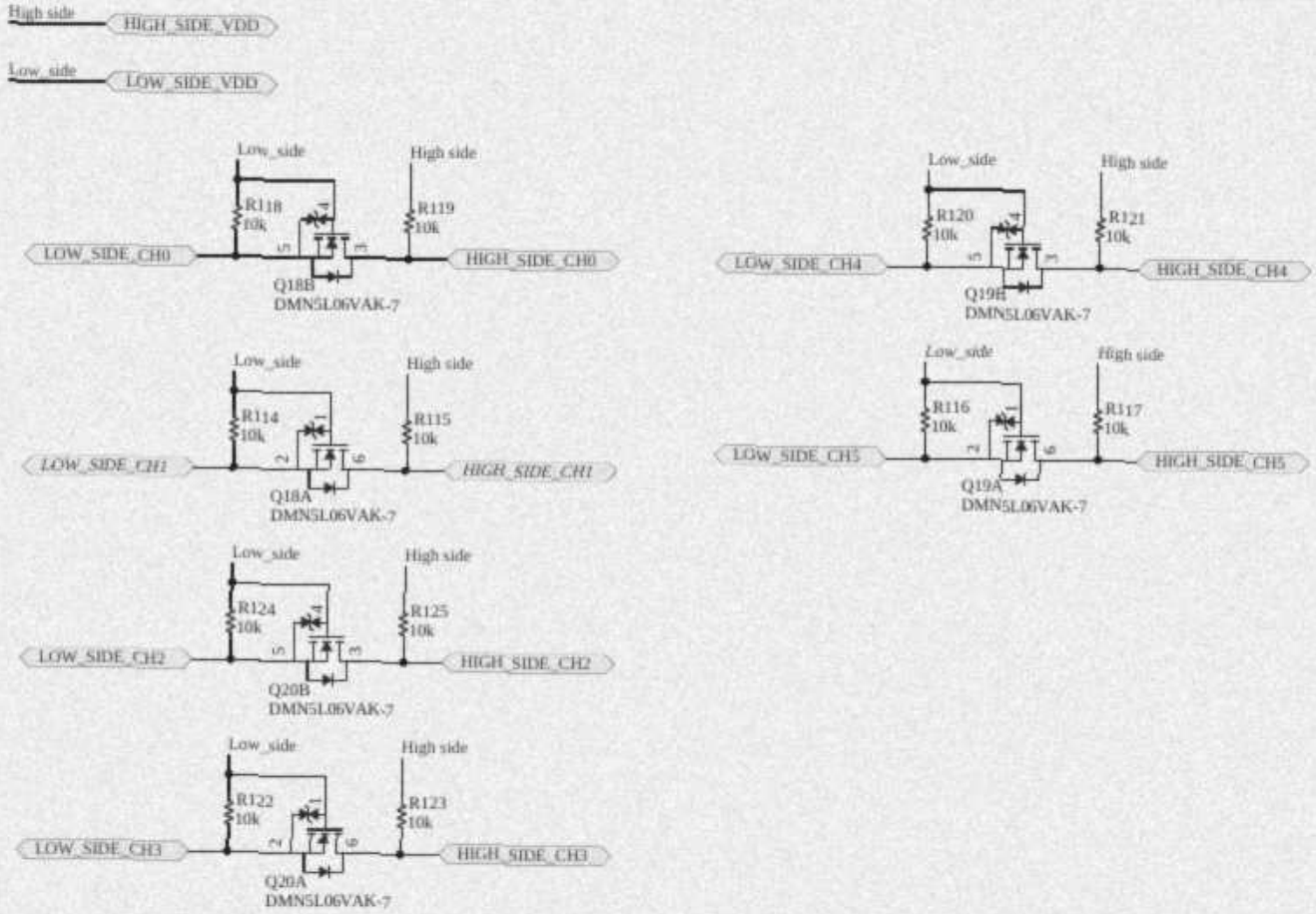
Title: TOP DIGITAL IN / ENCODERS			
ID: AKX00032		Revision: V1.1	
Date: 2023-05-11	Time: 15:11:06	Sheet 17 of 19	
File: IO_DIGITAL.SchDoc		Author: S.Navaretti	RevAuthor: S.Navaretti

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Title: DIGITAL IN			
ID: AKX00032		Revision: V3.1	
Date: 2021-05-11	Time: 11:11:07	Sheet: 7 of 19	
File: IO_DIGITAL_IN.SchDoc		Author: S.Navarotti	Reviewer: S.Navarotti



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Title: DIGITAL ENCODERS			
ID: AKX00032	Revision: V1.1		
Date: 2021-05-11 Time: 15:11:08	Sheet 19 of 19		
File: IO_DIGITAL_ENCODERS.SchDoc	Author: S.Navaretti	RevAuthor: S.Navaretti	

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